Week 7 — Simple regression I: estimation & interpretation

Business Forecasting

My code (.R) Notes

1 · A slope is an answer

Someone asks you how much a number moves when another number moves. That is the whole week.

The question on the table

You consult for Ecobici, the bike-share system in Mexico City.

Your client is planning for a warmer city. She has one question:

> When the daily temperature goes up by one degree, how many more trips do we get?

She does not want “more”. She wants a number, in trips per degree, because she has to size the fleet against it.

The data

`bikes` — 781 weekdays in Mexico City, 2017–2019.

Column	What it is
`date`	the day
`trips`	Ecobici trips that day
`temp`	average temperature, °C
`pm25`	fine-particle pollution
`weekday`	Monday … Friday
`friday`	1 if Friday, 0 otherwise

Look at it before you model it.

Exercise 1 — Meet the data

Print the first ten rows of `bikes`. Then find the mean and standard deviation of `bikes$trips` and of `bikes$temp`.

Exercise | Start Over | Show Hint | Run Code

```
# The first ten rows

# Centre and spread of trips

# Centre and spread of temp
```

Exercise 2 — Does temperature move trips?

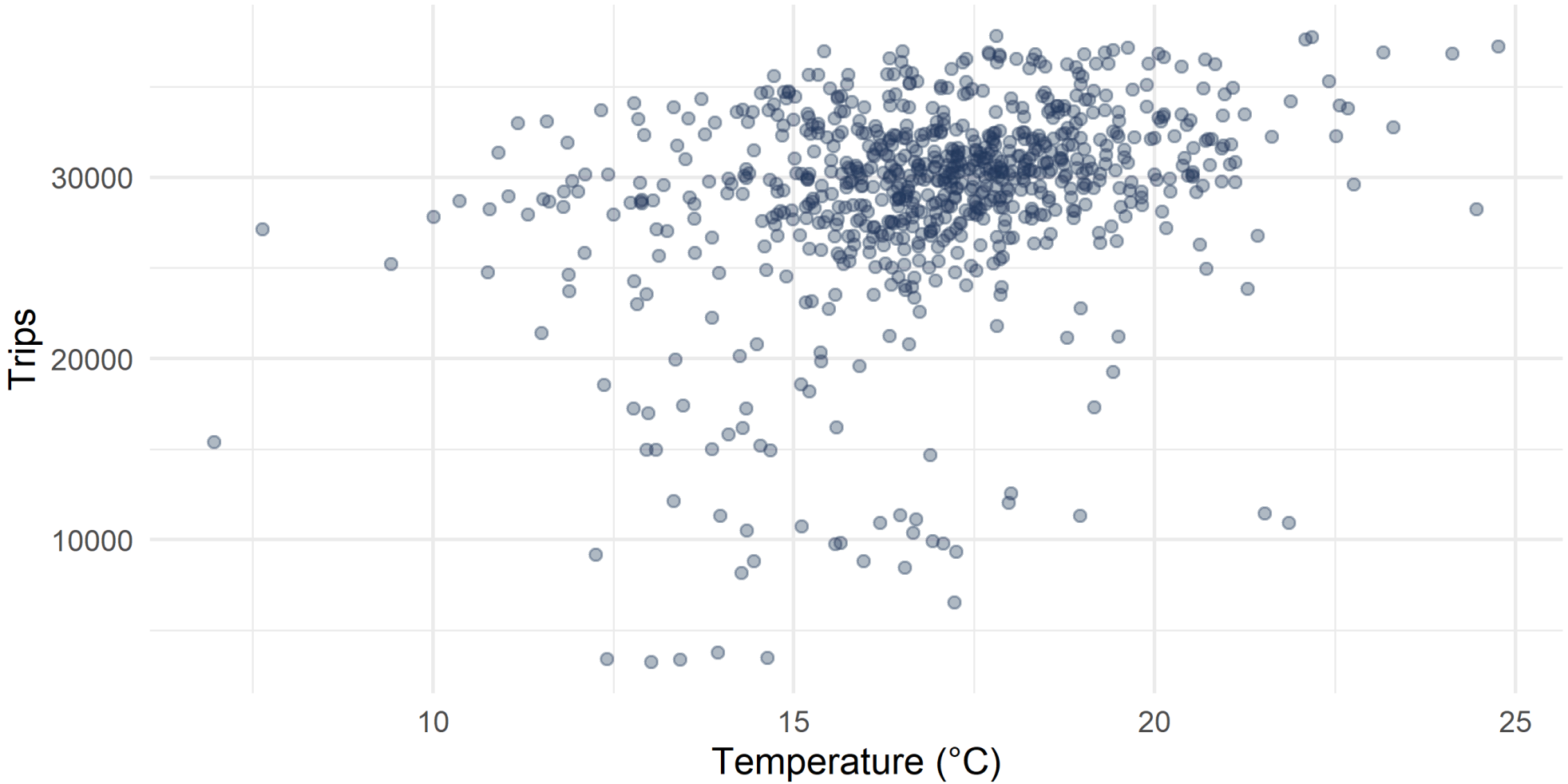
Draw the scatterplot of `bikes$trips` against `bikes$temp`, then compute the correlation between them.

Put temperature on the horizontal axis — it is the variable your client controls her planning around.

Exercise | Start Over | Show Hint | Run Code

```
1 # Scatterplot: temperature on x, trips on y
2
3 # Correlation
```

What you just drew



- A positive correlation tells you the cloud tilts up.
- It does not tell you how many trips per degree. That number is what the client needs.

Correlation cannot answer her

Correlation is unit-free. It is +0.297 whether you measure temperature in Celsius or Fahrenheit, trips in trips or in thousands of trips.

Your client needs an answer in trips per degree. That means putting a line through the cloud and reading its slope.

That line is the simple linear regression.

2 · The model

What we assume is true in the population, before we touch the data.

The population line

You have paired data $\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$.

We assume that in the population the two are linked by:

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

Reading the parts

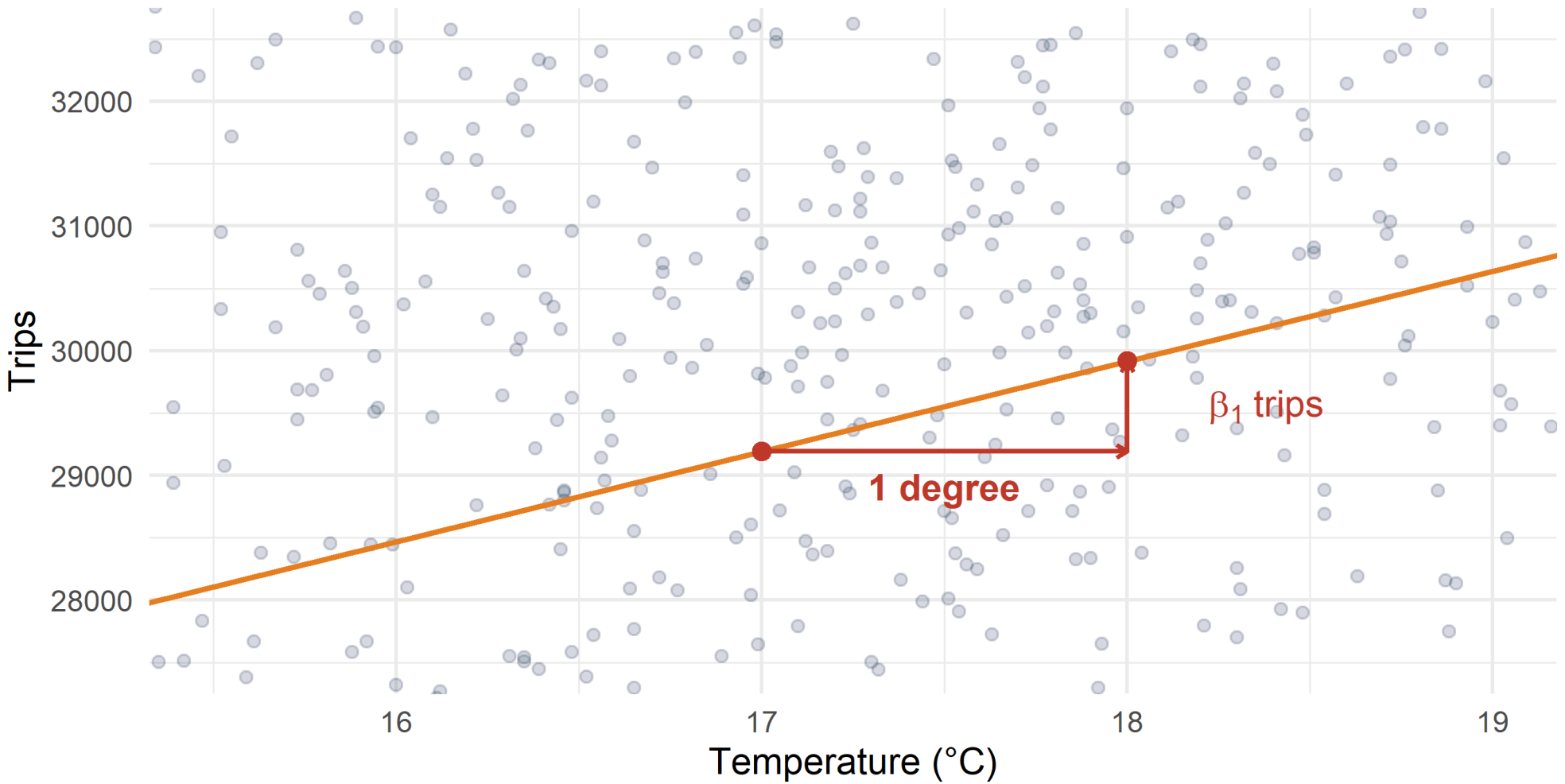
Symbol	Name	In the bike problem
y_i	dependent variable	trips on day i
x_i	regressor	temperature on day i
β_1	slope	extra trips per extra degree
β_0	intercept	trips at 0 °C
u_i	error	everything else

β_0 and β_1 are parameters. They are fixed, unknown numbers that describe the whole population. We never see them.

What we can do is estimate them from a sample. Those estimates get hats: $\hat{\beta}_0$ and $\hat{\beta}_1$.

Why the slope is the whole point

$$\beta_1 = \frac{\text{change in } y}{\text{change in } x}$$



- Step one degree along the line. The height you gain is β_1.
- That is the number the client sizes her fleet against.

The same question, five industries

In each row, what does the slope actually buy the business?

Setting	y	x	What β_1 buys you
Labour	wage	years of schooling	the return to one more year of education
Retail	units sold	price	how much demand falls when you raise the price
Real estate	sale price	square metres	what one more square metre is worth
Marketing	sales	ad spend	pesos of sales per peso of advertising
Pharma	prescriptions written	gifts to physicians	whether the gifts pay for themselves

Same model every time. Only the columns change.

3 · What we are assuming

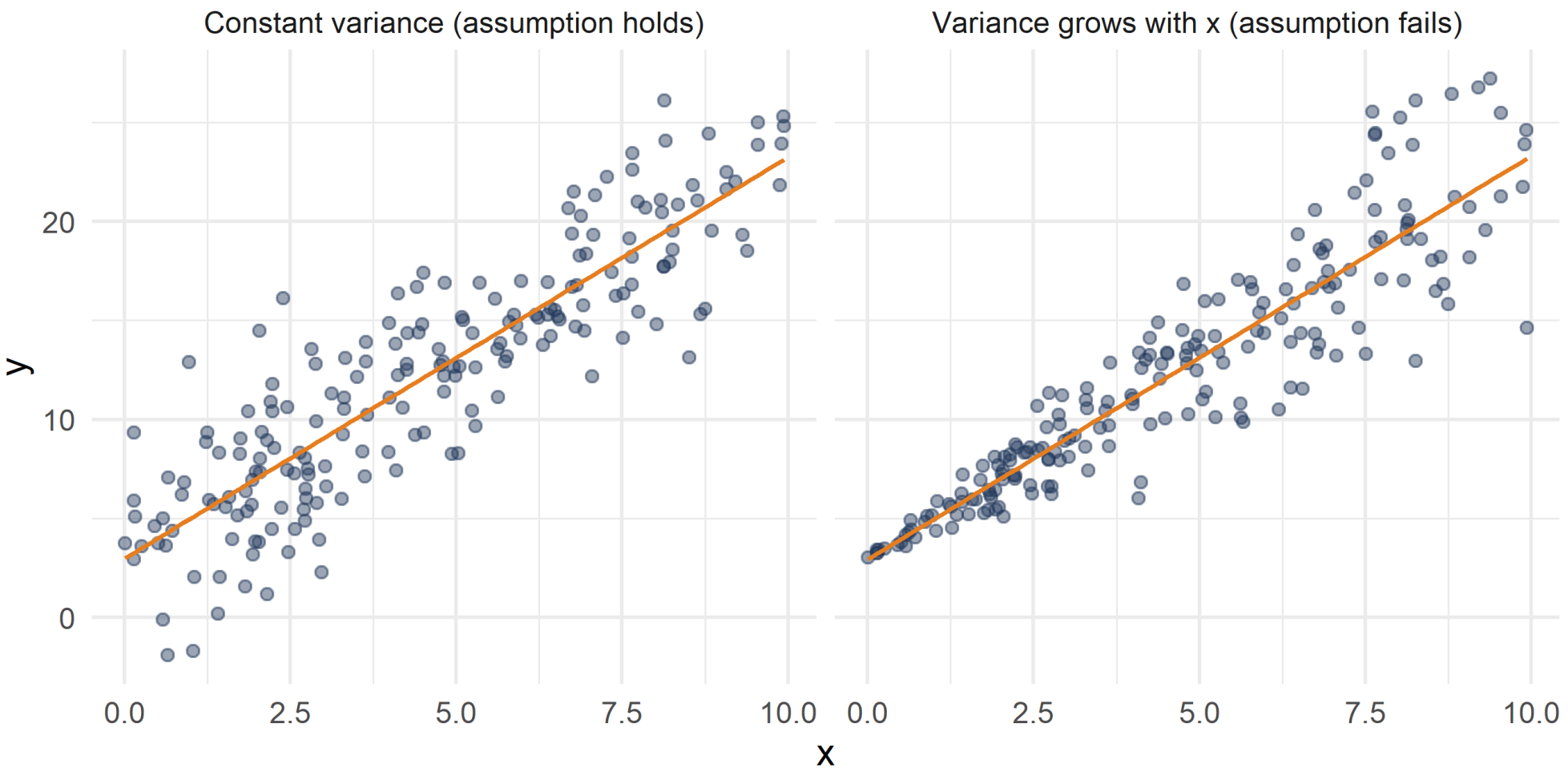
Four conditions. They decide whether the line means anything.

The four assumptions

1. Linear in the parameters, with an additive error. $y_i = \beta_0 + \beta_1 x_i + u_i$ — not β_1^2, not $\beta_0 \cdot u_i$.
2. The error has mean zero: $E(u_i) = 0$. On average the line is not too high or too low.
3. Constant variance: $\mathrm{Var}(u_i) = \sigma^2$. The scatter around the line is the same width everywhere. This is called homoskedasticity.
4. Errors are uncorrelated across observations: knowing one day's error tells you nothing about the next day's.

Assumptions 3 and 4 are the ones that fail in practice. W09 is about spotting that.

What constant variance looks like

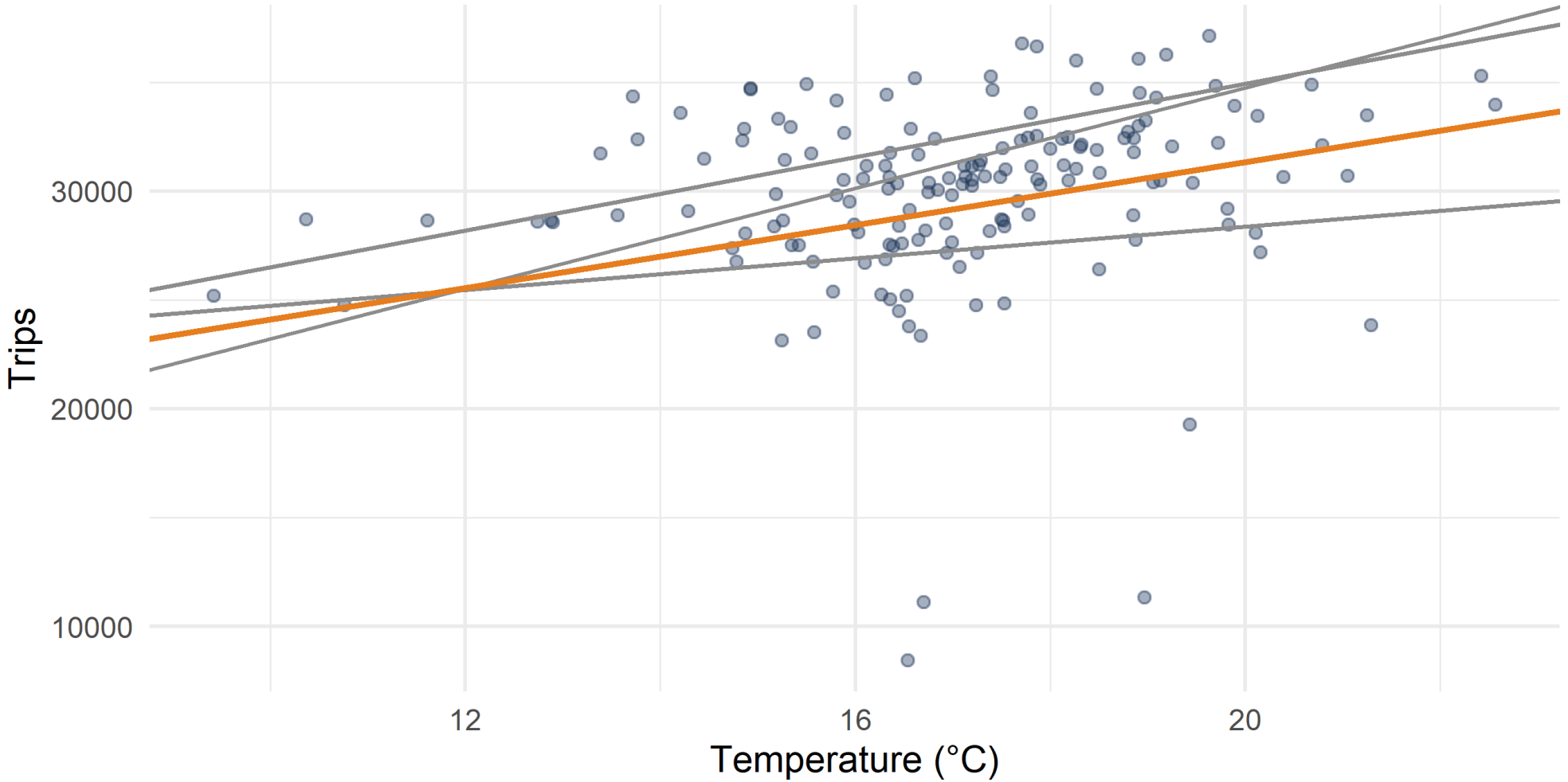


- Left: the band around the line has the same width at every x.
- Right: the band opens out. The line is still there, but how precisely we know it now depends on where you are.

4 · Fitting the line

Infinitely many lines pass through a cloud of points. One of them is best. Which, and why?

Which line?



- Every grey line is a candidate. So is the orange one.
- To pick, we need to say what best means.

Best means the residuals are small

For a candidate line with intercept $\hat{\beta}_0$ and slope $\hat{\beta}_1$, the fitted value for day i is what the line predicts:

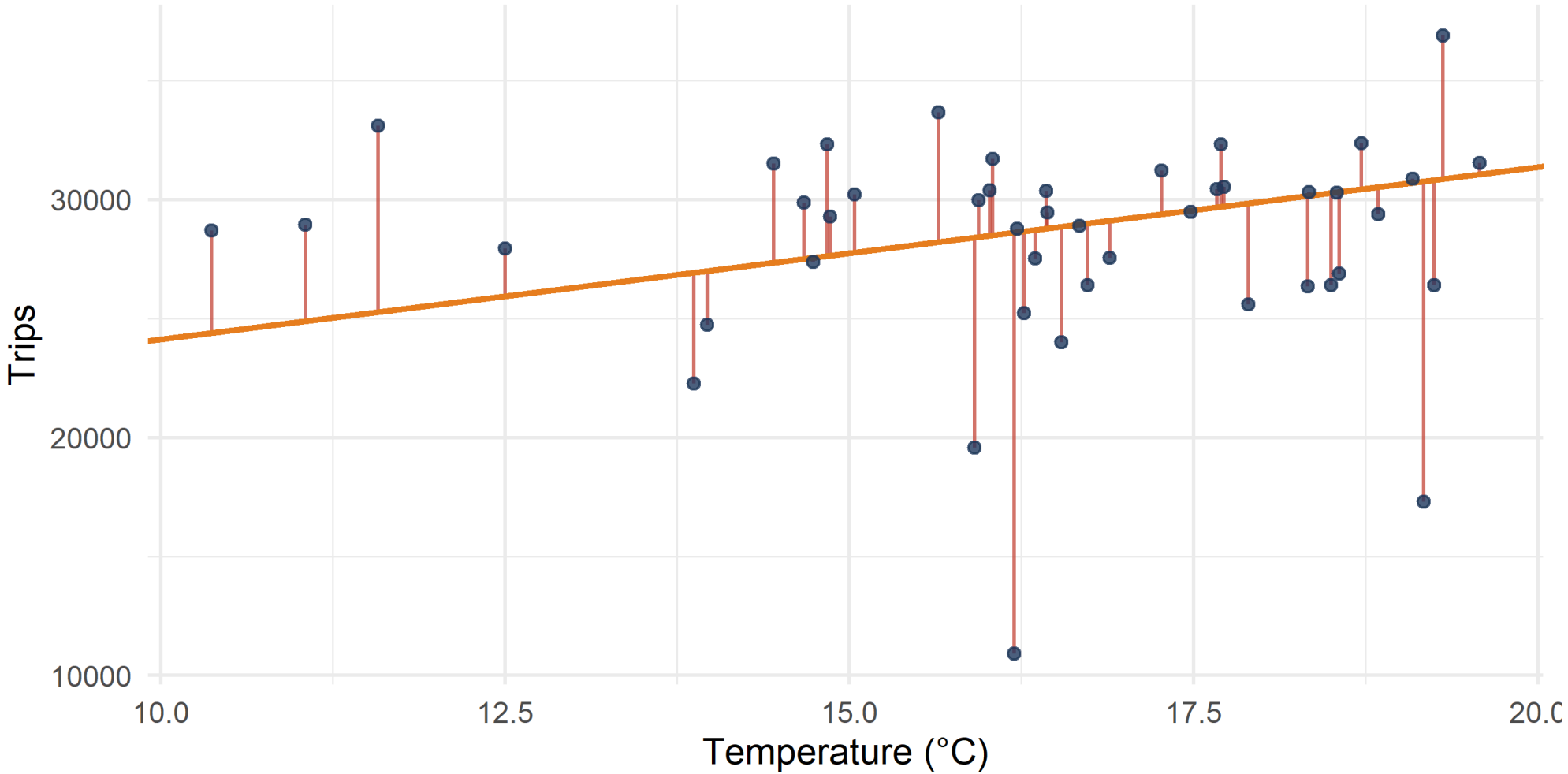
$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

The residual is how wrong the line was that day:

$$\hat{u}_i = y_i - \hat{y}_i$$

A residual is vertical distance: observed minus predicted. Positive means the day beat the line.

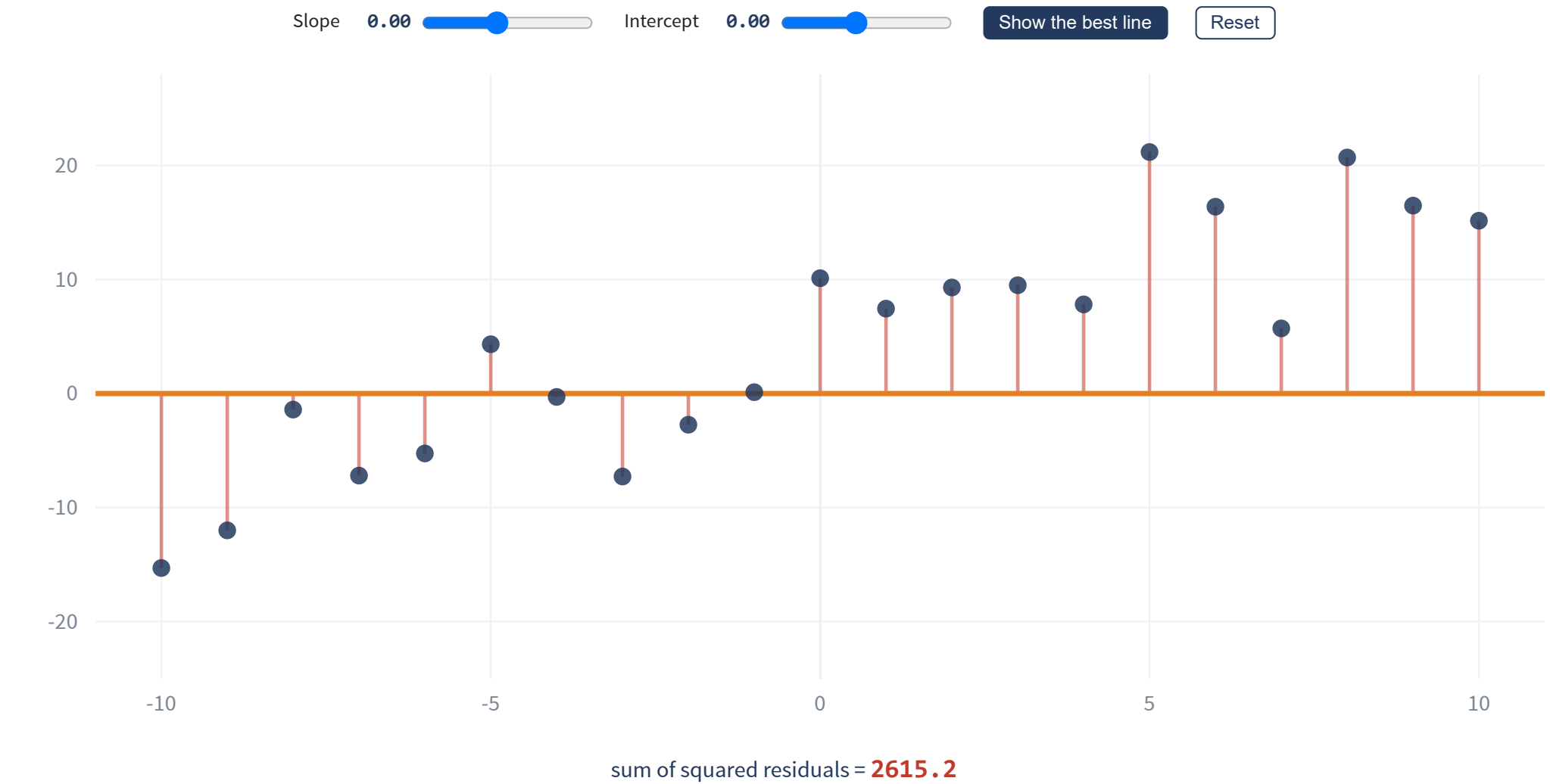
The residuals, drawn



- Each red segment is one residual.
- Move the line and every segment changes length.

Find the best line yourself

Move the two sliders. The red segments are the residuals; the number below is the sum of their squares. Get it as low as you can.



What the sliders were doing

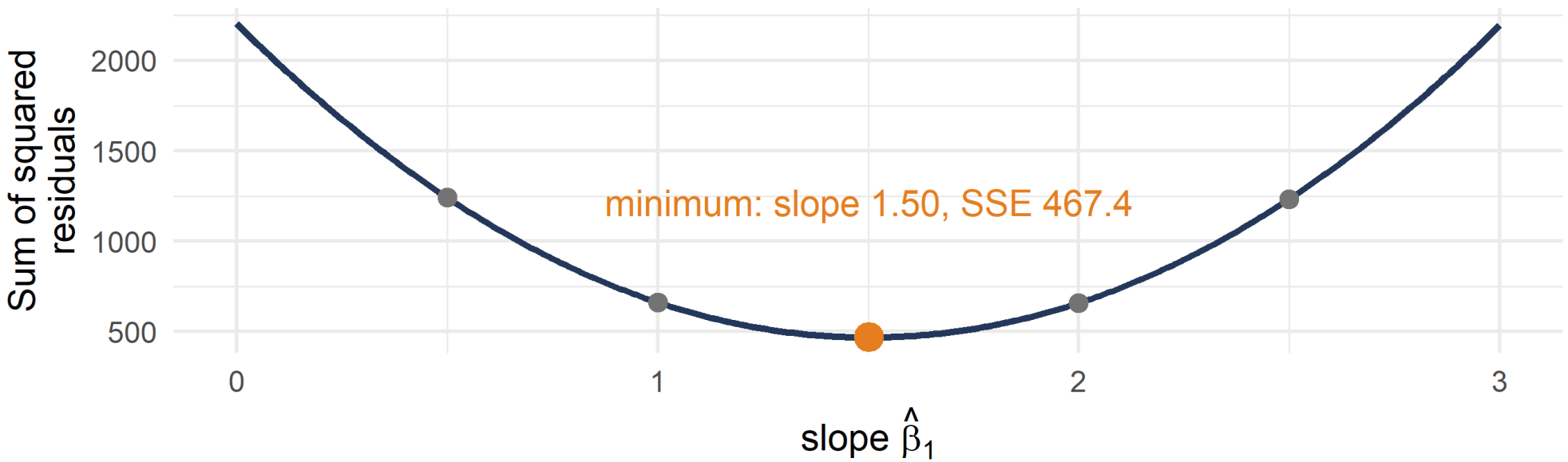
No matter how carefully you moved them, you could not get below 467.4. Every other line scores higher.

That minimum is not a coincidence, and you do not have to search for it — there is a formula that lands on it directly.

Ordinary least squares

We cannot just add the residuals — the positive and negative ones cancel. So we square them first, then add:

$$\min_{\hat{\beta}_0,\,\hat{\beta}_1} \sum_{i=1}^{n} \hat{u}_i^2 \;=\; \min_{\hat{\beta}_0,\,\hat{\beta}_1} \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2$$



The same 21 points as the sliders, with the intercept held at its best value (4.42). Each grey dot is one slope you could try. The sum is a parabola in $\hat{\beta}_1$, so it has exactly one lowest point.

This is ordinary least squares — OLS. The line that minimises this sum is the regression line.

The two estimators

Minimising gives closed-form answers. The slope:

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

And the intercept:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

The intercept formula says the line always passes through the point $(\bar{x}, \bar{y})$ — the centre of the cloud. See the algebra.

Two ways to say the same slope

Divide the top and bottom of $\hat{\beta}_1$ by $n - 1$ and it becomes a ratio you already know:

$$\hat{\beta}_1 = \frac{\mathrm{Cov}(x, y)}{\mathrm{Var}(x)}$$

Or, written with the correlation:

$$\hat{\beta}_1 = \mathrm{Cor}(x, y) \cdot \frac{\mathrm{sd}(y)}{\mathrm{sd}(x)}$$

The second form says it plainly: the slope is the correlation, converted into the units of the problem. That is exactly what your client needed and correlation alone could not give her.

Exercise 3 — Build the slope by hand

Compute $\hat{\beta}_1$ for `bikes$trips` on `bikes$temp` without using `lm`, using the covariance-over-variance formula. Then compute $\hat{\beta}_0$.

Exercise | Start Over | Show Hint | Run Code

```
1 # Slope: covariance of temp and trips, over the variance of temp
2
3 # Intercept: mean of trips, minus slope times mean of temp
```

✏️ Exercise 4 — Check it against the correlation form

Get the same slope a second way: correlation times the ratio of standard deviations. Do the two numbers agree?

Exercise | Start Over | Show Hint | ▷ Run Code

```
1 # Slope again: correlation of temp and trips, times sd(trips) / sd(temp)
```

Exercise 5 — Now let R do it

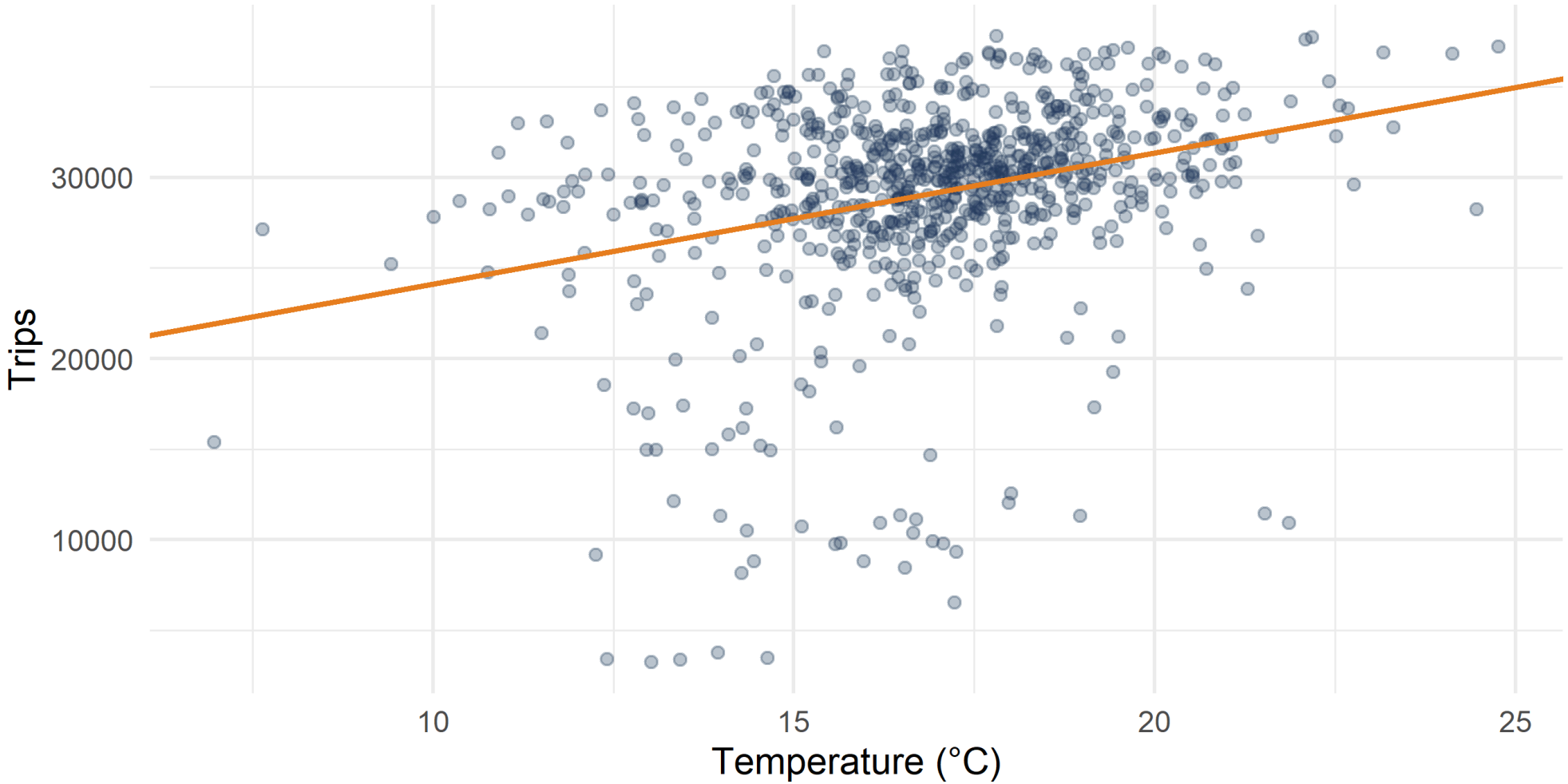
Fit the regression of `bikes$trips` on `bikes$temp` with `lm`, store it in `reg`, and print `reg`.

Exercise | Start Over | Show Hint | Run Code

```
# Fit the model and store it

# Print it
```

The fitted line



$$\widehat{\text{trips}} = 16{,}893 + 723.6 \times \text{temp}$$

The client’s answer

> A one-degree warmer day brings about 724 more trips.

And the intercept? It says a 0 °C day would see 16,893 trips. Mexico City never sees 0 °C — the coldest day here is 7 °C. The intercept is where the line *would* cross, not a prediction anyone should make.

Reporting an intercept far outside the range of your data is one of the most common ways to sound confident and be wrong.

Exercise 6 — Is the extra room worth building?

A landlord owns apartments in Mexico City. She can add 10 m^2 to one of them by joining a storage room to the flat. The work costs 150,000 pesos.

The data frame `rents` has 150 apartments she rents out:

- `rents$size` — floor area, in m^2
- `rents$rent` — monthly rent, in pesos

1. Regress `rents$rent` on `rents$size`. How many pesos of monthly rent does one extra m^2 bring, on average?
2. How much more rent per month should she expect from the extra 10 m^2?
3. How many months until the extra rent pays back the 150,000 pesos?
4. What does the intercept say? Should she use it?

Exercise | Start Over | Show Hint | Run Code

```
1
```

5 · Reading the output

`summary()` prints eight things. This week we need two of them.

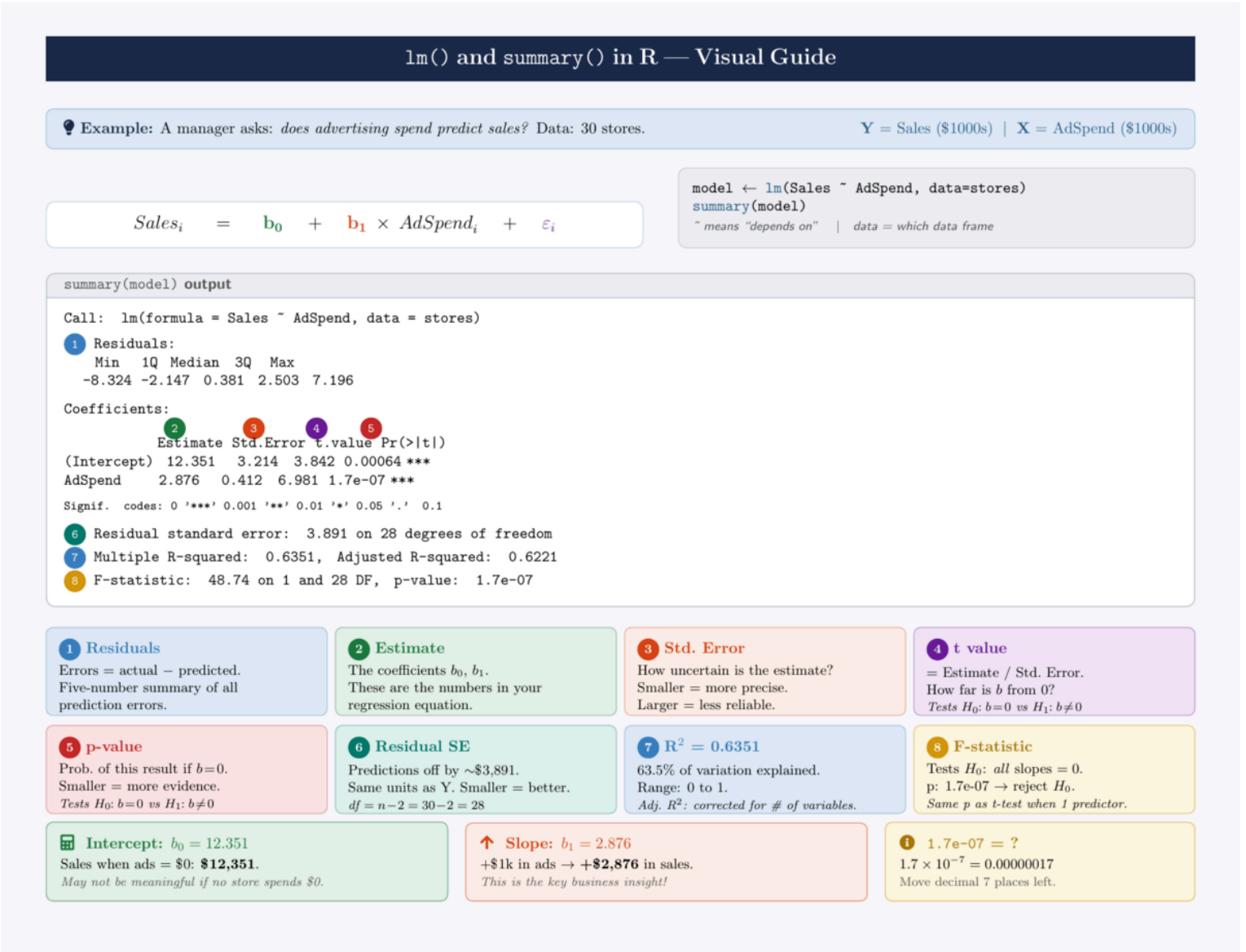
Exercise 7 — Look at the full output

Run `summary()` on your fitted model. Find the two coefficients you computed by hand.

Exercise | Start Over | Show Hint | Run Code

```
1 reg <- lm(trips ~ temp, data = bikes)
2
3 # Show the full summary
```

What each piece is



This week and next week

This week you read the `Estimate` column and the `R-squared` line. That is estimation and interpretation.

Next week the other three columns — `Std. Error`, `t value`, `Pr(>|t|)` — become the subject. They answer a different question: *given that the slope is 723.6 in this sample, how sure are we that it is not zero in the population?*

Do not read them yet. Build them first — that is W08.

The full output, if you want to look ahead

Fitted values and residuals

Two things the model hands back for every day in the data:

```
fitted(reg)      # what the line predicted, one number per day
residuals(reg)   # observed minus predicted, one number per day
```

Both are on the exam sheet, and both are vectors as long as the data.

Exercise 8 — Was the line high or low?

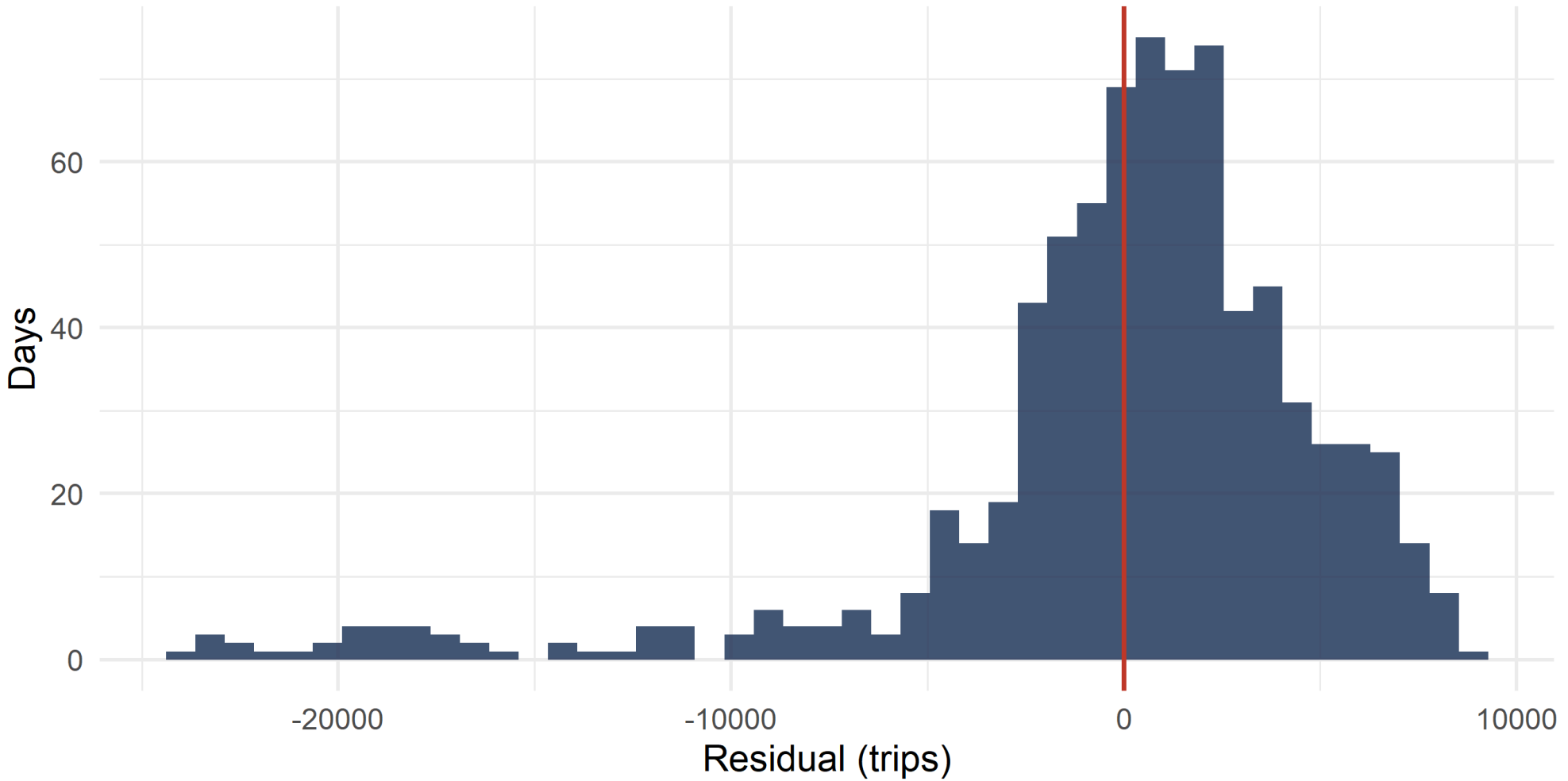
Using your fitted `reg`:

1. What share of days did the line under-predict — that is, what share of residuals are positive?
2. Draw a histogram of the residuals. Are they roughly symmetric around zero?

Exercise | Start Over | Show Hint | Run Code

```
1 reg <- lm(trips ~ temp, data = bikes)
2
3 # Share of residuals above zero
4
5 # Histogram of residuals
```

The residuals, seen

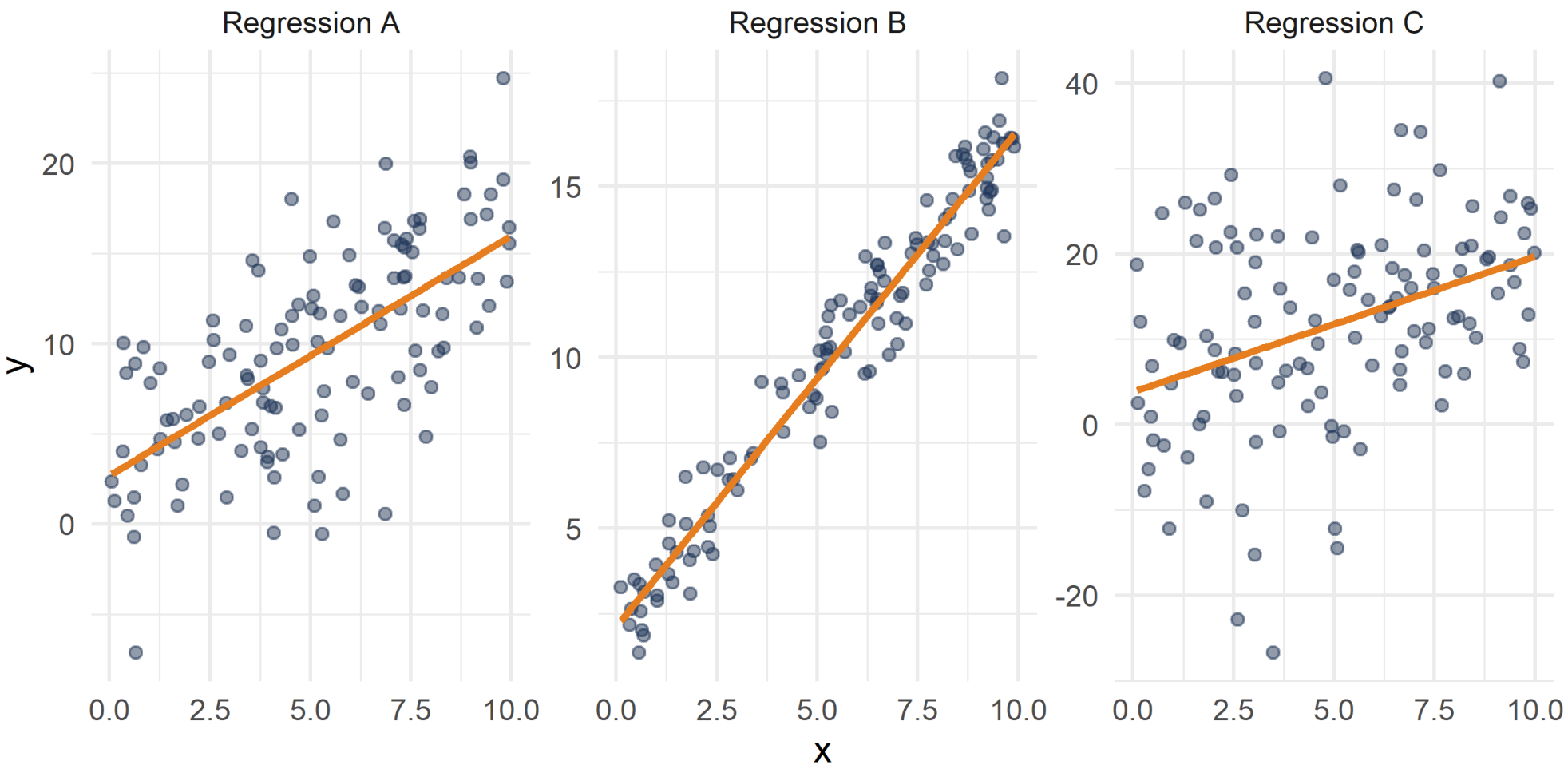


- Most days sit within a few thousand trips of the line.
- The left tail is long. A model built on temperature alone cannot explain a holiday.

6 · How much does it explain?

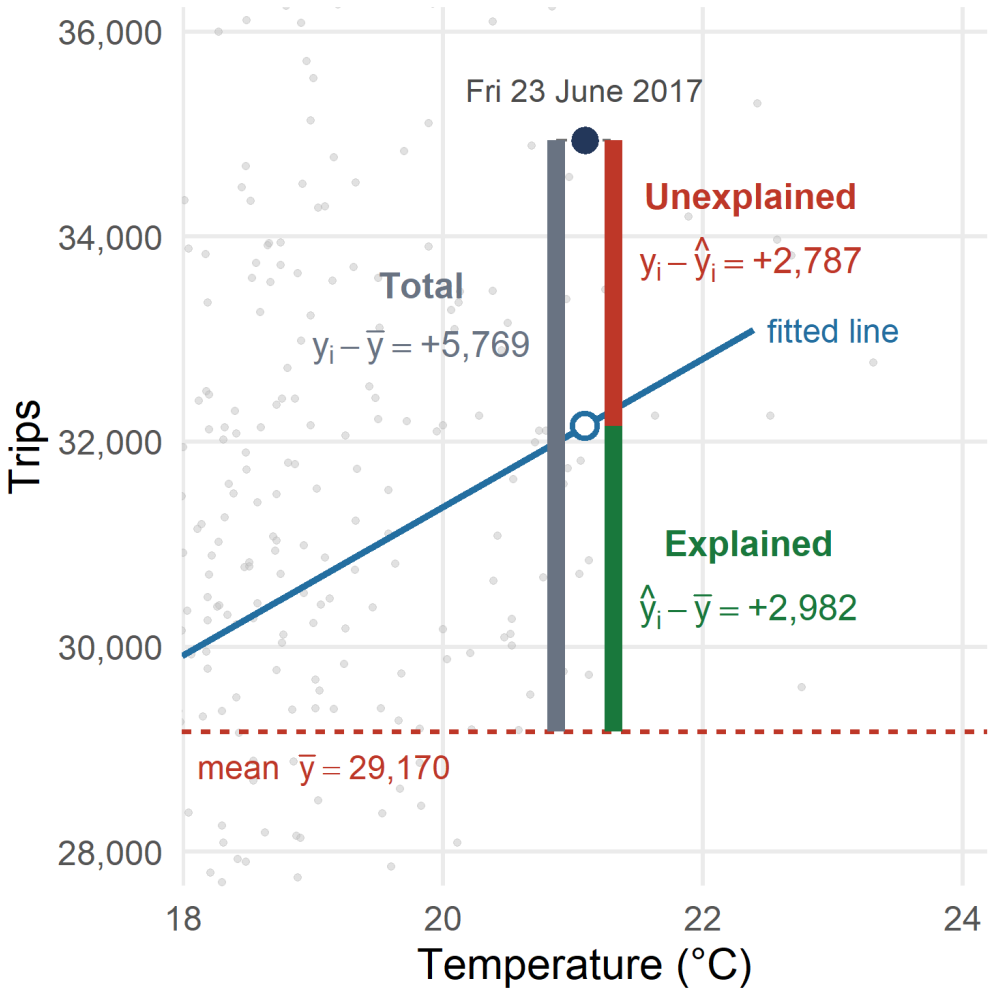
A slope can be real and still leave most of the story untold.

Three regressions



How do we know which regression is the best at making predictions?

Splitting the variation



For one day i, the distance from the mean splits in two:

$$\underbrace{y_i - \bar{y}}_{\text{total}} = \underbrace{\hat{y}_i - \bar{y}}_{\text{explained}} + \underbrace{y_i - \hat{y}_i}_{\text{unexplained}}$$

Square the pieces and add them over all days. Total variation in y splits the same way:

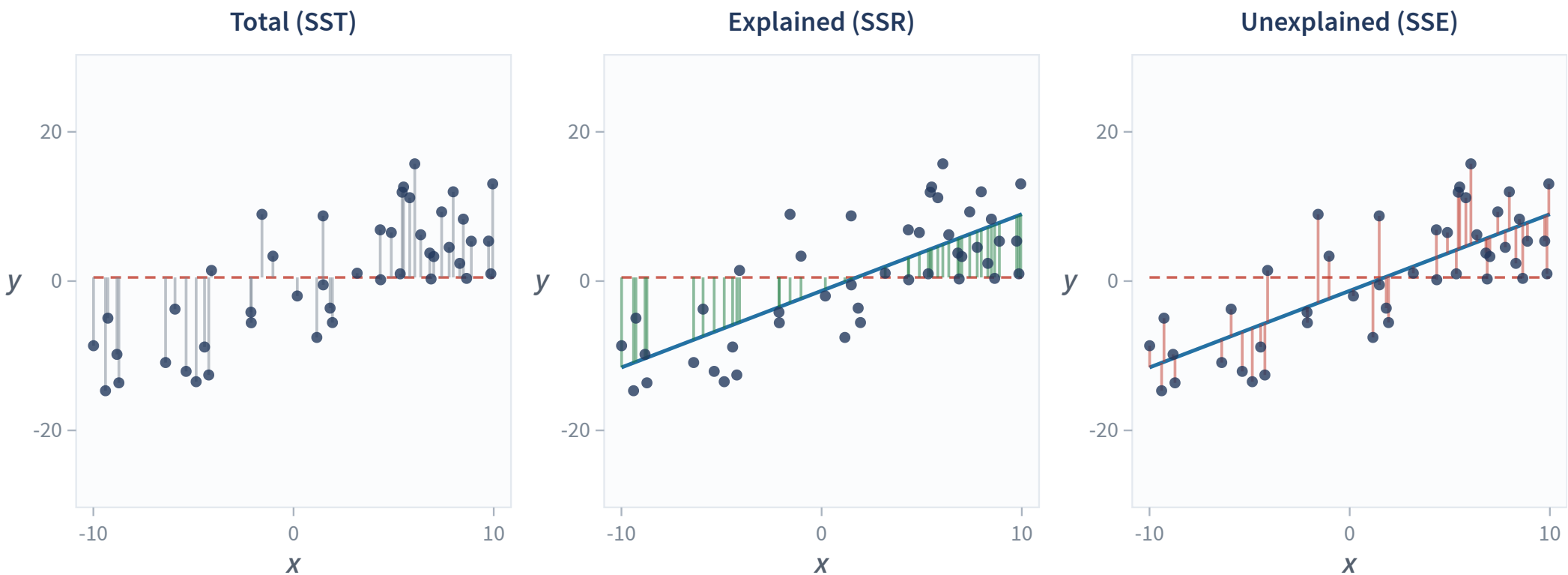
$$\underbrace{\sum (y_i - \bar{y})^2}_{SST} = \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{SSR} + \underbrace{\sum (y_i - \hat{y}_i)^2}_{SSE}$$

Every bit of variation in y is either explained by the line or left in the residuals. Nothing goes missing.

Watch the variation split

Left: how far each point sits from the mean. Middle: how much of that the line accounts for. Right: what is left over. Dashed red line: the mean $\bar{y}$. Blue line: the fitted line.

Slope 1.00 Noise 5.0 New sample Show the numbers



SST = ? = SSR ? + SSE ? → R^2 = ?

What the three panels showed

- Flatten the slope and the middle panel empties — the line explains nothing, and $SSR \to 0$.
- Turn the noise down and the right panel empties — nothing is left over, and $SSE \to 0$.
- The left panel is the job. The other two always add up to it.

R-squared

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

- The share of the variation in y that the line accounts for.
- Always between 0 and 1.
- With a single regressor, $R^2 = [\text{Cor}(x, y)]^2$.

`anova(reg)` prints the pieces directly — that is what it is for.

Exercise 9 — Decompose the variation

Run `anova(reg)`. Read `SSR` and `SSE` out of the `Sum Sq` column, then build R^2 from them.

Exercise | Start Over | Show Hint | Run Code

```
reg <- lm(trips ~ temp, data = bikes)

# The decomposition table

# Pull the two sums of squares out and form R-squared
```

Exercise 10 — The correlation shortcut

With one regressor, R^2 is the squared correlation. Check it: square the correlation between `bikes$temp` and `bikes$trips` and compare with what you just got.

Exercise | Start Over | Show Hint | Run Code

```
1 # Square the correlation
```

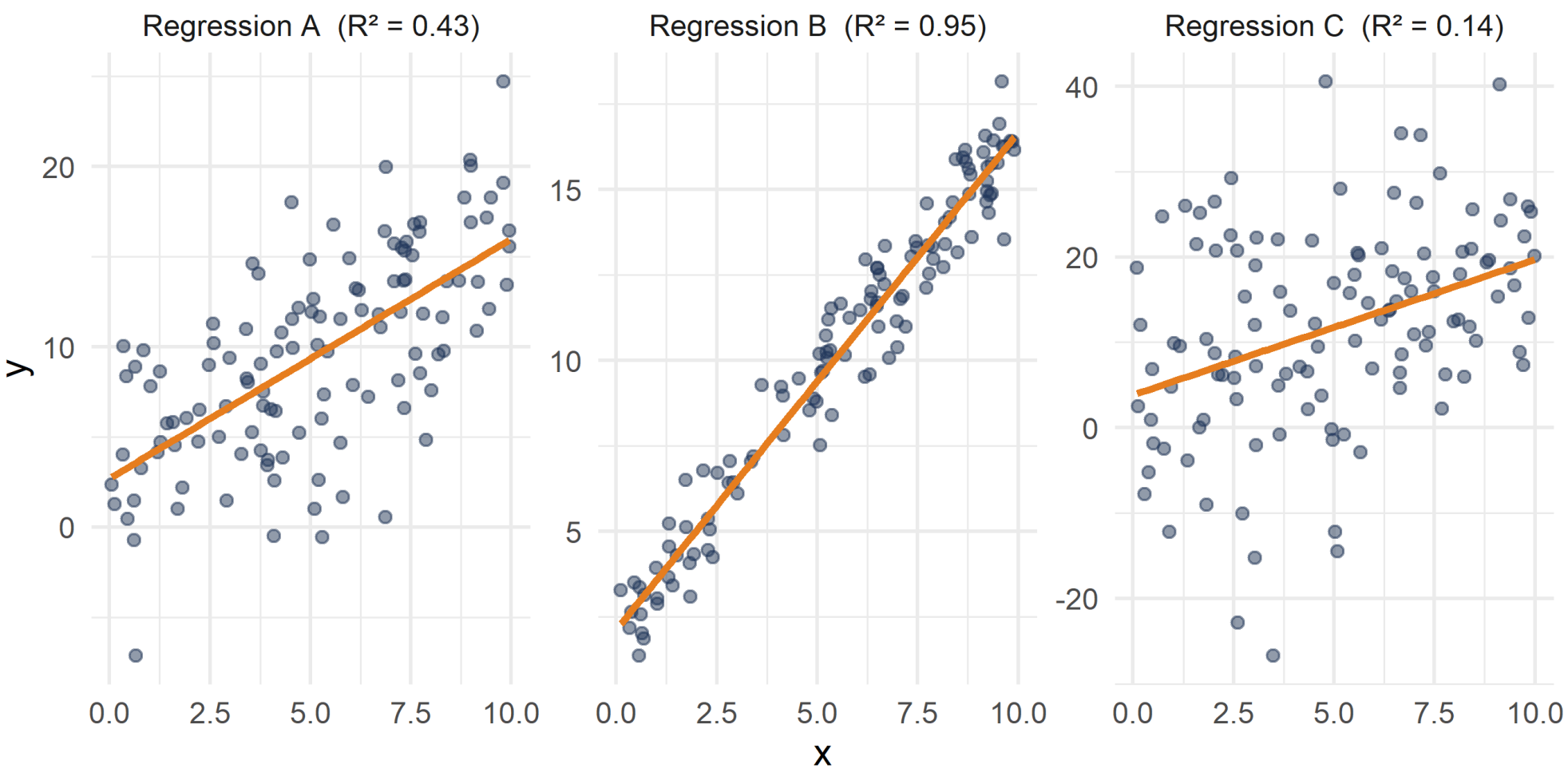
8.8%

Temperature explains under nine percent of the day-to-day variation in Ecobici trips.

The slope is still real and still useful: 724 trips per degree is what the client plans against. But 91% of what moves ridership is something else — the day of the week, rain, holidays, a metro closure.

A low R^2 does not mean the slope is wrong. It means the line is not the whole story. Those are different claims, and confusing them is a classic exam trap.

What different R^2 looks like



All three have the same slope. Only the scatter differs.

Back to the opening question: regression B predicts best. Its points sit closest to its line, and its R^2 is the highest.

🖊️ Exercise 11 — How much of the rent does size explain?

Back to the landlord and `rents` (`rents$size` in m^2, `rents$rent` in pesos a month). Before she trusts the extra 1,785 pesos, she wants to know how well size alone explains rent.

1. Find SSR, SSE and SST for the regression of `rents$rent` on `rents$size`.
2. Compute R^2. What share of the variation in rents does size explain?
3. Which of the three panels on the previous slide does this look most like? Compare it with the bikes regression, where $R^2 = 0.088$.

Exercise | Start Over | Show Hint | Run Code

```
1
```

7 · Changing the units

Your client reports in thousands. Finance wants yearly cost, not monthly salary. Does the answer change?

The problem

You estimated a regression in pesos per hour of training. You now have to present it to a client in New York who thinks in dollars.

Do you have to re-estimate? Or can you convert the coefficients you already have?

What happens to the slope

Rescale x to $z = ax + c$ and y to $y' = by + d$. Then:

$$\hat{\beta}_1' = \text{Cor}(z, y') \cdot \frac{\text{sd}(y')}{\text{sd}(z)} = \text{Cor}(x, y) \cdot \frac{b \cdot \text{sd}(y)}{a \cdot \text{sd}(x)} = \frac{b}{a} \hat{\beta}_1$$

Three things fall out:

- correlation does not change when you rescale — so it drops out;
- adding a constant to either variable leaves the slope alone;
- multiplying changes it by exactly the ratio b/a.

What happens to the intercept

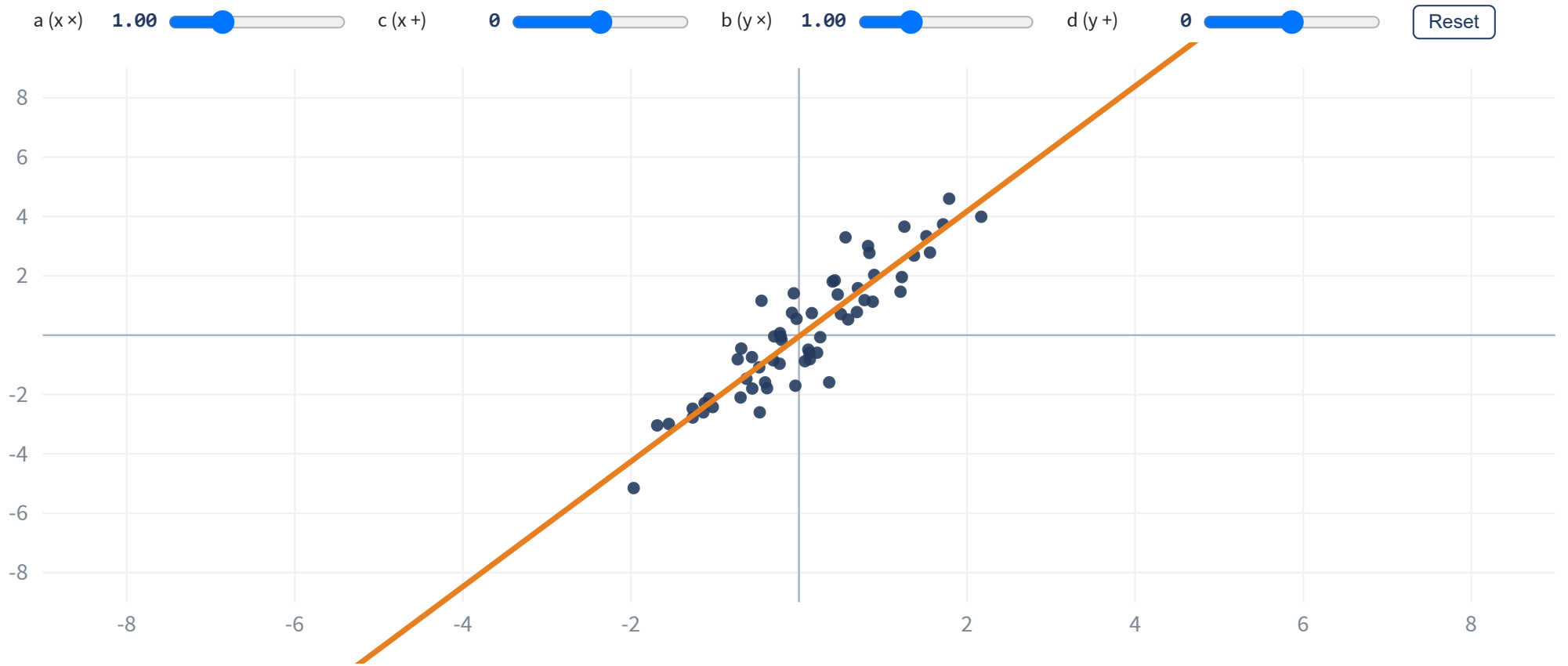
$$\hat{\beta}_0' = b\,\hat{\beta}_0 + d - \frac{b}{a}\,\hat{\beta}_1 c$$

- multiplying y changes it;
- adding a constant to y changes it;
- adding a constant to x changes it;
- multiplying x only matters if you also shifted x.

Full derivation in the appendix.

Rescale the axes and watch

Grey is the original data, black is the rescaled data. The dashed line is the original fit, the solid line is fitted to the rescaled data.



original b_0 =-0.04 b_1 = 2.11 | rescaled b_0 =-0.04 b_1 = 2.11

predicted by the formulas: $(b/a) \cdot b_1 = 2.11$ $b \cdot b_0 + d - (b/a) \cdot b_1 \cdot c = -0.04$

What to try on it

- Set $b = 2$, leave everything else — the slope doubles, the cloud stretches vertically.
- Set $a = 2$, leave everything else — the slope halves.
- Move c or d alone — the cloud slides, the slope does not move at all.
- Set $a = 2$ and $c = 3$ — now the intercept moves too. That is the $-\frac{b}{a}\hat{\beta}_1 c$ term earning its place.

The green line under the plot is the formula's prediction. It never disagrees with the refitted number.

Exercise 12 — Trips in thousands

Your client wants trips reported in thousands. Make a new column `bikes$trips_k` equal to `bikes$trips` divided by 1000, regress it on `bikes$temp`, and compare the slope with the original 723.55.

Predict the answer before you run it.

Exercise | Start Over | Show Hint | Run Code

```
1  # Trips in thousands
2
3  # Regress the new variable on temperature
```

Exercise 13 — What does a hire really cost?

A tech company in Guadalajara is planning next year's hiring budget. The data frame `hires` has its 120 engineers:

- `hires$experience` — years of experience
- `hires$salary` — monthly gross salary, in pesos

Finance does not budget in monthly salary. They budget the first-year cost of a hire: 12 months of salary, plus 35% on top for IMSS, aguinaldo and benefits, plus 25,000 pesos once for recruiting and a laptop:

$$\text{cost} = 12 \times 1.35 \times \text{salary} + 25{,}000 = 16.2 \times \text{salary} + 25{,}000$$

1. Regress `hires$salary` on `hires$experience`. What do the slope and the intercept say?
2. Use the rescaling rules to predict the slope and the intercept of cost on experience. Do this before you run anything.
3. Build `hires$cost`, regress it on `hires$experience`, and check your prediction.
4. How much does one extra year of experience add to the first-year cost of a hire?

Exercise | Start Over | Show Hint | Run Code

```
1
```

8 · A dummy regressor

What if x is not a number, but a yes-or-no?

Binary variables

A dummy (or binary) variable takes only two values:

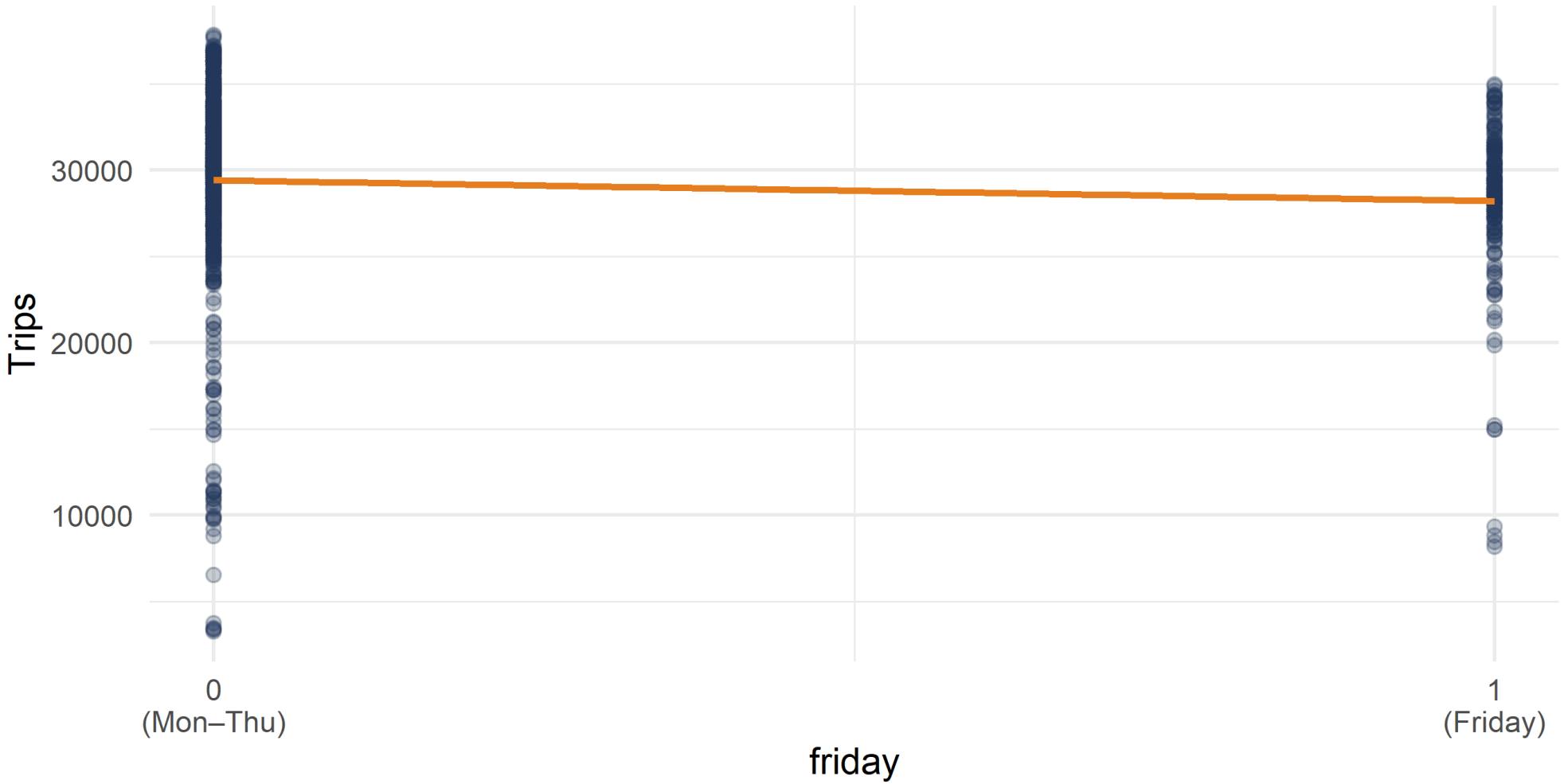
$$x_i = \begin{cases} 1 & \text{if the condition is true} \\ 0 & \text{otherwise} \end{cases}$$

Examples:

- $x_i = 1$ if customer i is female, 0 if male
- $x_i = 1$ if transaction i is fraudulent, 0 otherwise
- $x_i = 1$ if client i made a purchase, 0 if not

A dummy on the x-axis

Our bike data again: `bikes$friday` is 1 on Fridays and 0 on the other weekdays.



- Every day sits at $x = 0$ or $x = 1$, so the dots form two columns.
- The orange line is the ordinary least-squares fit, the same `lm()` as before.

Where does the line cross each column?

What the coefficients mean now

Regress y_i on a dummy x_i. The estimates collapse to something very simple:

$$\hat{\beta}_0 = \bar{y}_{x=0} \qquad \text{the mean of } y \text{ in the } 0 \text{ group}$$

$$\hat{\beta}_1 = \bar{y}_{x=1} - \bar{y}_{x=0} \qquad \text{the difference in group means}$$

Which makes sense: "a one-unit increase in x" is the *only* move available — it means switching from the 0 group to the 1 group.

Exercise 14 — Fridays

`bikes$friday` is 1 on Fridays and 0 on other weekdays.

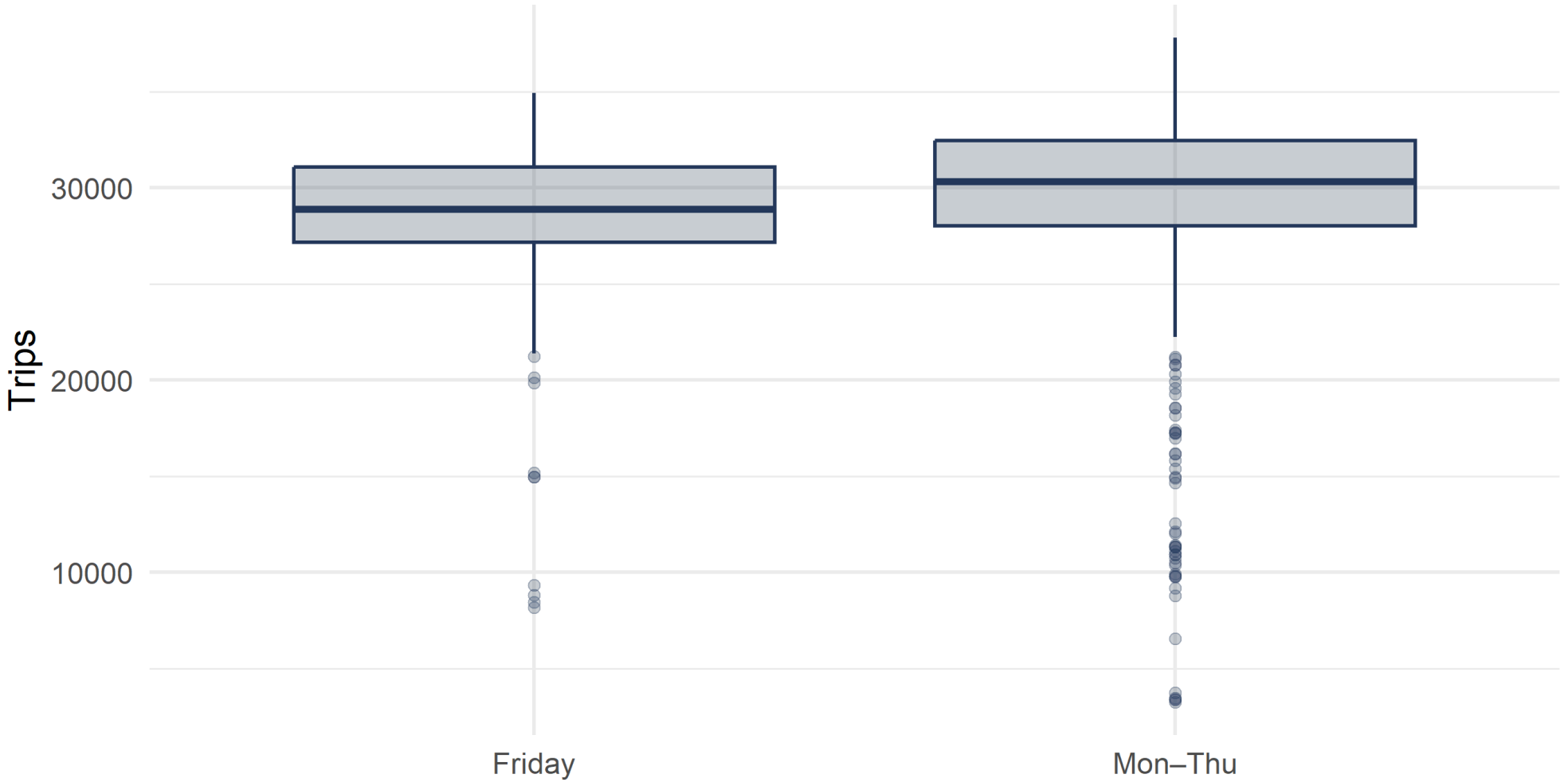
1. Compute the mean of `bikes$trips` on Fridays and on non-Fridays separately.
2. Then regress `bikes$trips` on `bikes$friday`.

Before you run step 2: what do you expect the intercept and the slope to be?

Exercise | Start Over | Show Hint | Run Code

```
1 # Mean trips on non-Fridays
2
3 # Mean trips on Fridays
4
5 # Now the regression
```

Fridays are quieter



- About 1,200 fewer trips on a Friday.
- Whether that gap is big enough to act on is a question about uncertainty — W08.

Exercise 15 — The pay gap

An HR consultancy is checking pay in Mexico. The data frame `salaries` has 20,000 workers from the Mexican labour force survey (ENOE, 2022):

- `salaries$income` — monthly income, in pesos
- `salaries$hours_worked` — hours worked last week
- `salaries$female` — 1 for women, 0 for men

1. Regress `salaries$income` on `salaries$female`. What is the mean monthly income of men? By how many pesos is the mean income of women lower?
2. Regress `salaries$hours_worked` on `salaries$female`. What are the mean weekly hours of men? How many fewer hours do women work, on average?
3. Suppose women earned the same pesos per hour as men, for their own mean hours. What would their mean monthly income be? How much of the monthly gap would be left? What does this tell the consultancy about where the gap comes from?

Exercise | Start Over | Show Hint | Run Code

```
1
```

9 · A binary outcome

Now flip it. What if y is the yes-or-no?

The setup

You work at Amazon. You want to predict whether a customer returns a product, from the product's star rating.

$$y_i = \begin{cases} 1 & \text{if customer } i \text{ returns the product} \\ 0 & \text{if they keep it} \end{cases}$$

And you run the ordinary regression anyway:

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

where x_i is the product's average star rating, from 1 to 5.

The data frame `returns` has 500 orders:

- `returns$rating` — the product's average star rating
- `returns$returned` — 1 if the customer returned it, 0 if they kept it

Exercise 16 — Plot the returns

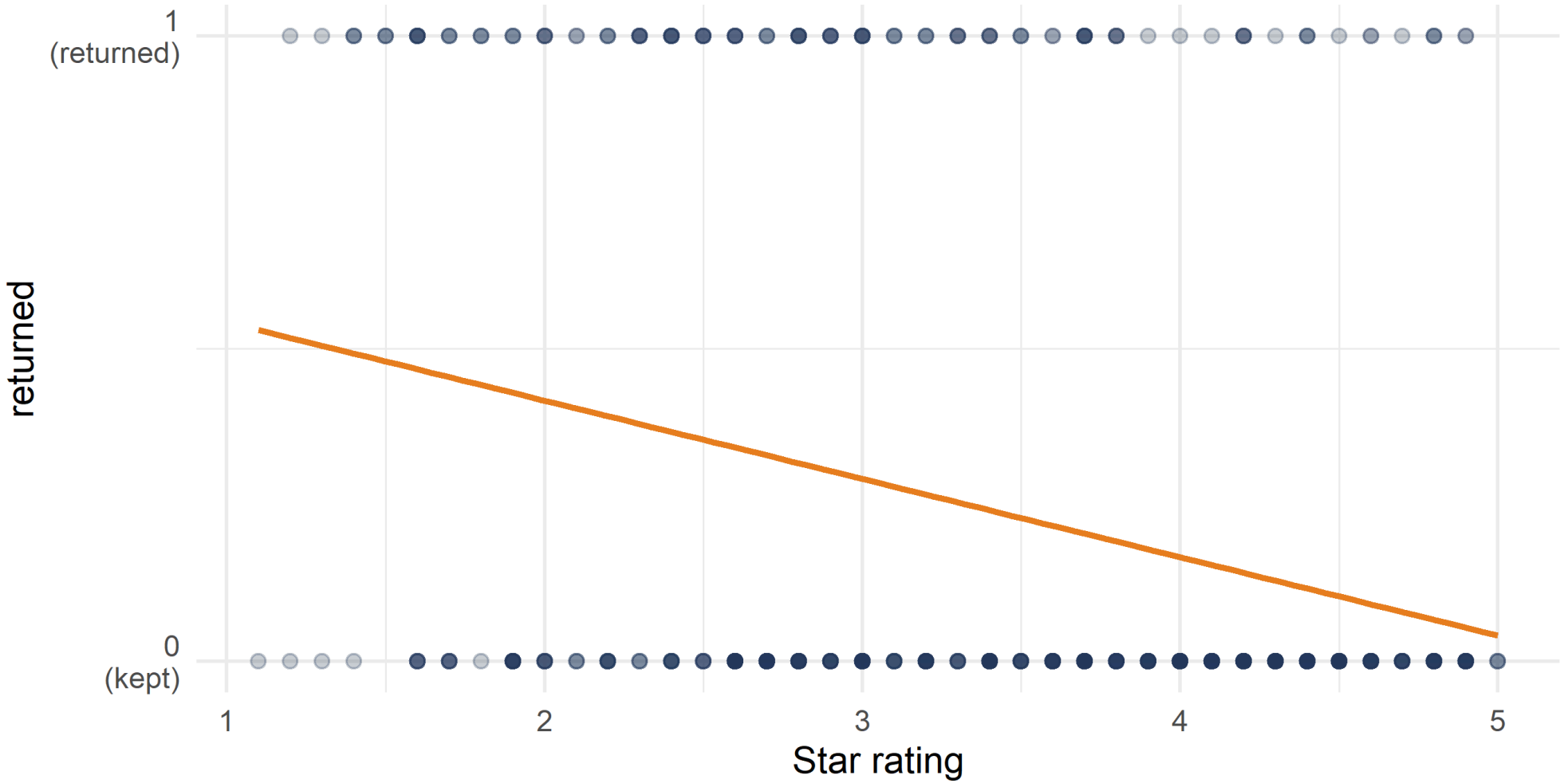
Draw a scatterplot with `returns$rating` on the x-axis and `returns$returned` on the y-axis.

What does a scatterplot look like when y can only be 0 or 1?

Exercise | Start Over | Show Hint | Run Code

```
1 # Scatterplot: rating on x, returned on y
```

Two lines of dots



- The top line thins out as the rating rises. The bottom line fills up.
- The orange line is the ordinary least-squares fit.

A straight line through 0s and 1s. Does it mean anything?

Why it works

A regression predicts the expected value of y given x:

$$E(y \mid x) = \beta_0 + \beta_1 x$$

The expected value of a variable is its mean. And for a 0/1 variable, the mean is just the proportion of 1s.

So $E(y \mid x)$ is the share of products with rating x that get returned — which is exactly the probability of return at that rating.

This is called the linear probability model. The orange line is a probability of return, read off at each rating.

Exercise 17 — Run the regression

Regress `returns$returned` on `returns$rating` and print the full output.

1. What is the estimated slope? Say what it means in words.
2. What is the predicted probability of a return for a 4-star product? For a 5-star product?

Exercise | Start Over | Show Hint | Run Code

```
1 # Fit the model and store it
2
3 # Show the full summary
4
5 # Predicted probability at 4 stars and at 5 stars
```

The results

	Estimate	Std. error	t value
Intercept	0.667	0.067	9.89
rating	-0.125	0.019	-6.53

$n = 500, R^2 = 0.079.$

The fitted model:

$$\widehat{P}(\text{return} \mid x) = 0.667 \ - 0.125\, x$$

Reading the results formally

- Slope, $\hat{\beta}_1 = -0.125$: a one-star increase in a product's rating is associated with a 12.5 percentage point decrease in the probability that the customer returns it.
- Intercept, $\hat{\beta}_0 = 0.667$: the predicted probability of a return for a product rated 0 stars. No product has 0 stars (the lowest here is 1.1), so this is an extrapolation with no business meaning on its own.
- Fitted values: at 4 stars, $0.667 - 0.125 \times 4 = 0.166$, a 16.6% chance of a return. At 5 stars, 4.1%.
- $R^2 = 0.079$: the rating explains about 8% of the variation in returns. Most of what makes a customer send a product back is not in the rating.

What the slope means

β_1 is the change in the probability of $y = 1$ when x rises by one unit.

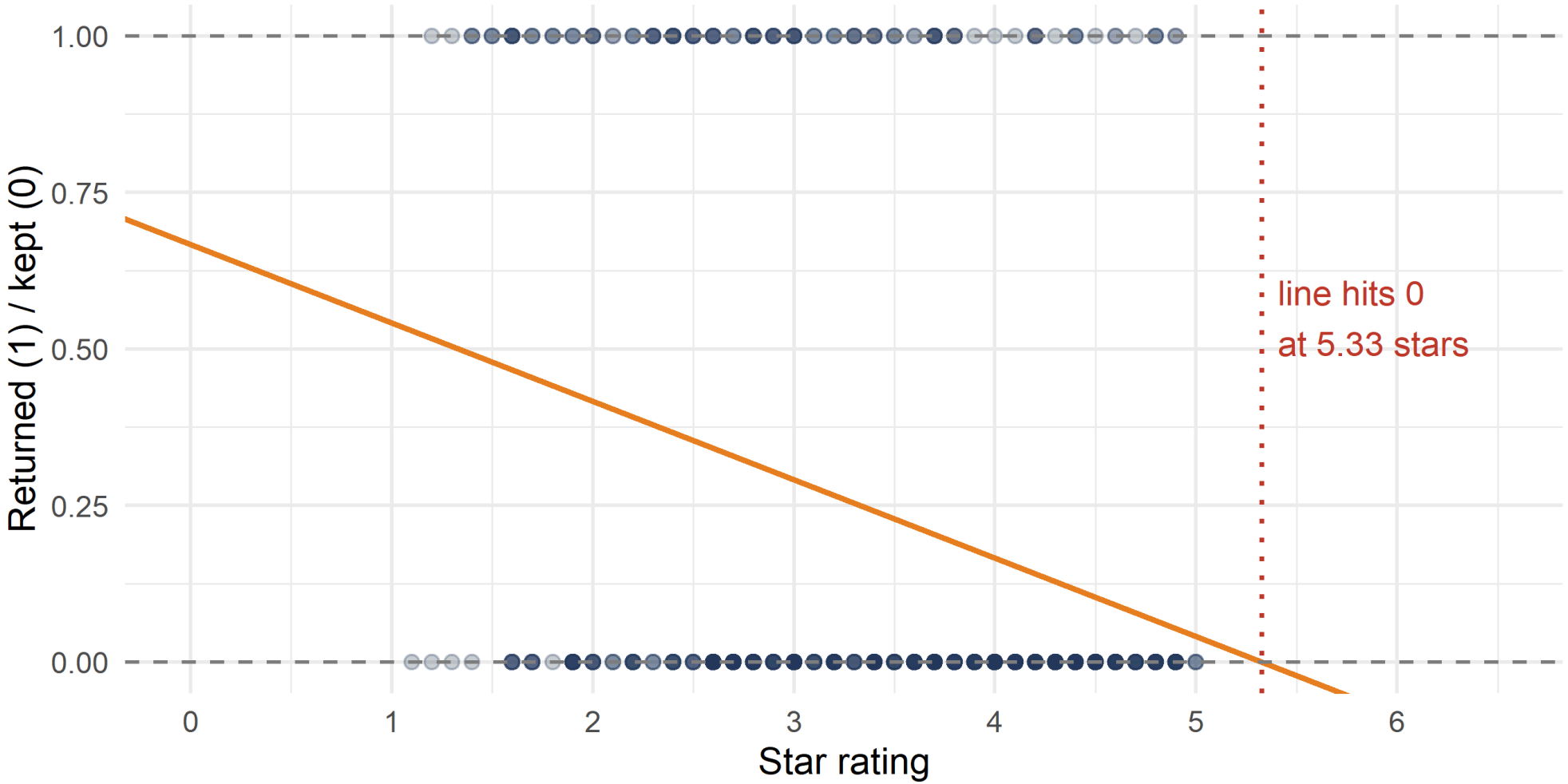
From 4 stars to 5 stars, the return probability goes from 16.6% to 4.1%:

$$\hat{\beta}_1 = 0.041 - 0.166 = -0.125$$

Each extra star cuts the chance of a return by 12.5 percentage points.

Say *percentage points*, not *percent*. Going from 16.6% to 4.1% is a fall of 12.5 percentage points, and also a fall of about 75 percent. Those are different numbers and graders notice.

Where it breaks



- The line is straight, so it will eventually leave the 0–1 band.
- Here it crosses zero at 5.33 stars. Past that, it would predict a negative probability of return, which is not a probability.

Worth it anyway

The limitation: OLS can predict outside $[0, 1]$, which is nonsense for a probability. In practice people move to logit or probit models, which are non-linear and stay inside the band.

Why we still teach it: the coefficient reads directly as a change in probability, in percentage points. A logit coefficient does not. For a first look at a binary outcome, the linear probability model is simple and honest about what it is.

Exercise 18 — Pricing health insurance

A health insurer sets its prices by age. Clients with diabetes cost more to cover. The data frame `health` has 5,000 adults from a health survey:

- `health$age` — age in years
- `health$diabetes` — 1 if the person has diabetes, 0 if not

1. Regress `health$diabetes` on `health$age`. By how many percentage points does the probability of diabetes rise with each extra year of age? With ten extra years?
2. What is the predicted probability of diabetes for a 30-year-old client? For a 60-year-old client? Should the insurer charge them the same price?
3. What is the predicted probability for a 20-year-old client? Why can the insurer not use this number?

Exercise | Start Over | Show Hint | Run Code

```
1
```

What you should be able to do

- Write the population model and say what each of β_0, β_1, u_i means.
- State the four assumptions, and recognise constant variance in a picture.
- Compute $\hat{\beta}_1$ and $\hat{\beta}_0$ by hand, two ways, and with `lm`.
- Read the `Estimate` column and the R^2 off `summary()`.
- Split SST into SSR and SSE with `anova()` and build R^2.
- Say what a low R^2 does and does not mean.
- Convert coefficients when the units of x or y change.
- Interpret a dummy regressor as a difference in group means.
- Interpret a fitted value from a binary outcome as a probability, in percentage points.

Next week: the same regression, but asking how sure we are — standard errors, t-statistics, p-values, confidence intervals and prediction intervals.

Go to the exam practice →

Practice for this week

Weekly exercise set: `W07_exercises.pdf`, with solutions in `W07_solutions.pdf`.

You can already solve exercises 1–5 and 7–13.

Not in these slides: exercise 6, when a slope is a causal effect.

Suggested before the next class, closest to the exam:

Exercise	Topic	How
13	A line by hand and with `lm`, ENOE	R with the data
3	Out-of-pocket medical costs and age (`hrs`)	R with the data
4	Weekly earnings and education (`cps`)	R with the data
5	Child IQ and parents' IQ: three regressions	R as a calculator
2	Tortilla machinery: a dummy regressor	R as a calculator

Exam practice

Problems from this week's set. ← Back to the summary

Exam practice 1 — Standardised variables

P2 #1

Take a sample $\{(x_i, y_i)\}_{i=1}^{n}$ where x and y are both standardised — each has mean 0 and standard deviation 1. Their sample correlation is $r_{xy} = 0.5$.

For the model $y_i = \alpha + \beta x_i + e_i$, what are the least-squares estimates $\hat{\alpha}$ and $\hat{\beta}$?

Work it from the formulas first. Then build a small standardised sample in the cell and confirm.

Exercise | Start Over | Show Hint | Run Code

```
1 # Build x and y with mean 0, sd 1 and correlation 0.5, then regress
```

Exam practice 2 — Over or under?

P2 #9

A regression predicts the shelf life of an apple from its weight. For one apple the predicted shelf life is 4.6 days and the residual is −0.6 days.

Did the model over-estimate or under-estimate that apple's shelf life? What was the apple's actual shelf life?

Exercise | Start Over | Show Hint | Run Code

```
1 # Residual = observed - fitted. Solve for the observed value.
```

Exam practice 3 — Out-of-pocket medical costs

P2 #3

`hrs` holds 6,052 non-married individuals aged 50+. `outofpocket_costs` is annual out-of-pocket medical spending in 2000; `age` is age in years; `male` is 1 for men.

1. Fit the model with `lm` and store it in `hrs_results`.
2. Interpret the slope.
3. What is the estimated difference in expected out-of-pocket costs between a 70-year-old and a 60-year-old?
4. What share of fitted values are negative? Is that a problem, given costs cannot be below zero?
5. Fit the model separately for men and women. Compare the slopes.

Exercise | Start Over | Show Hint | Run Code

```
# 1. Fit and store

# 3. Ten years of age

# 4. Share of fitted values below zero

# 5. Men and women separately
```

Exam practice 4 — Reading a fit

P2 #7, P2 #8

Answer both from the definitions — no data needed.

1. A regression predicts weight (kg) from height (cm) for adult males. Would you expect the slope to be positive or negative? Would you expect R^2 to be close to 1?
2. The uncertainty in the slope estimate $\hat{\beta}_1$ is higher when there is
    i. a lot of scatter around the line, or (II) very little scatter. Which, and why?

Exercise | Start Over | Show Hint | Run Code

```
1 # Simulate it: same slope, two different amounts of scatter.
2 # Fit both and compare how far each estimated slope lands from the truth.
```

Appendix — optional

The algebra behind the results in the main flow.

Appendix — deriving the OLS estimators

We minimise

$$S(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i \right)^2$$

Take the derivative with respect to $\hat{\beta}_0$ and set it to zero:

$$\frac{\partial S}{\partial \hat{\beta}_0} = -2 \sum \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i \right) = 0$$

Divide by $-2n$:

$$\bar{y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{x} = 0 \quad \Longrightarrow \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

That is the intercept, and it shows the line runs through $(\bar{x}, \bar{y})$.

Now the slope:

$$\frac{\partial S}{\partial \hat{\beta}_1} = -2 \sum x_i \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i \right) = 0$$

Substitute $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ and rearrange:

Appendix — the intercept under rescaling

With $z = ax + c$ and $y' = by + d$, we already have $\hat{\beta}_1' = (b/a)\hat{\beta}_1$. Then:

$$\begin{aligned}
\hat{\beta}_0' &= \bar{y}' - \hat{\beta}_1' \bar{z} \\
&= (b\bar{y} + d) - \left(\frac{b}{a}\hat{\beta}_1\right)(a\bar{x} + c) \\
&= b\bar{y} + d - b\hat{\beta}_1\bar{x} - \frac{b}{a}\hat{\beta}_1 c \\
&= b\left(\bar{y} - \hat{\beta}_1\bar{x}\right) + d - \frac{b}{a}\hat{\beta}_1 c \\
&= b\,\hat{\beta}_0 + d - \frac{b}{a}\hat{\beta}_1 c
\end{aligned}$$

← Back to What happens to the intercept

Appendix — regression through the origin

Suppose we force the line through zero:

$$y_i = \beta_1 x_i + u_i$$

Minimising $\sum (y_i - \hat{\beta}_1 x_i)^2$ gives

$$\hat{\beta}_1 = \frac{\sum x_i y_i}{\sum x_i^2}$$

Note this is not the usual estimator — the deviations from means are gone.

If the true line does *not* pass through the origin and you use this anyway, the slope absorbs the missing intercept and is biased. Forcing a line through zero is a strong claim about the world, not a simplification.

Appendix — the full `summary()` output

```r
reg <- lm(trips ~ temp, data = bikes)
summary(reg)
```

```
Call:
lm(formula = trips ~ temp, data = bikes)

Residuals:
     Min       1Q   Median       3Q      Max
-24010.5  -1508.4    774.5   2920.5   8900.2

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 16892.66    1427.32  11.835   <2e-16 ***
temp          723.55      83.37   8.679   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5302 on 779 degrees of freedom
Multiple R-squared:  0.08817,   Adjusted R-squared:  0.087
F-statistic: 75.32 on 1 and 779 DF,  p-value: < 2.2e-16
```

Columns two, three and four are W08.

← Back to This week and next week